

CPR BLRSC

SHORT CIRCUIT LABORATORY

2084

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Log Sheet for temperature - rise test on Oil immersed / Dry type transformers

Tes. Series Number	25T1654		Sample Number	HVD2550847	
Name of the Customer / Manufacturer					
Rating		kVA: 250		No. of Phases	
Voltage Class		HV 11kV LV 433V		Rated HV Current	
IS : 1880		IEC : —		Rated LV Current	
In Accordance with		Other : —		Customers Instructions: —	
Condition of Sample :		4022		Test Date (s): 9 & 10/11/25	

Source Utilized for Temperature Rise Test		Single Phase Source		Three Phase Source	
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Test Method Employed : Short Circuit Method		No Load Run (In case Dry Type)		Load Run (in Case of Dry type)	
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Connections	HV (or) LV	Connected to 3Phase AC Supply	NLL (W)	FLL (W)	Total Loss
	HV	Shorted through copper links	383.26	2239.14	2628.2

Oil Cooled : Total loss injection (No load loss+ Full load loss at 75 °C up to Steady state)= 2628.20W					
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Dry Type : Applying Rated No- load Voltage on LV — or HV — side					
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Number of Taps : 7		Tap Position for TR Test : LTG		Current at Selected Tap : 14.58A	
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HV winding Resistance : 3.5540 Ω or mΩ		LV winding Resistance : 3.8844 mΩ or uΩ		Ambient during Cold Resistance Measurement : 25.9 °C	
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Readings during Temperature Rise Test

Hr	Time of starting test (AM/PM)	Applied Total Losses NLV / FLG	Temperature on °C for Steady state prediction		Ambient Temperature (°C)				Average Ambient Temp (°C)	Top oil temperature Rise (°C)
			Top oil	Bottom oil	1	2	3	4		
0	9:40	2623.2	23.10	21.65	21.10	21.00	21.30			
1	10:40	—	29.80	23.25	20.80	20.65	21.15			
2	11:40	—	34.55	26.00	20.65	20.45	21.05			
3	12:40	—	38.80	28.55	20.45	20.25	20.85			
4	01:40	—	41.55	30.50	20.35	20.05	20.75			
5	02:40	—	43.45	31.80	20.35	20.00	20.75			
6	03:40	—	44.00	32.20	20.20	19.80	20.60			
7	04:40	—	45.50	33.00	20.10	19.60	20.50			
8	5:40	—	46.00	33.60	20.00	19.50	20.40	—	19.96	26.03
9	6:40	—	46.55	34.05	19.80	19.35	20.30	—	19.81	26.73
10	7:40	—	46.90	35.50	20.10	19.75	20.45	—	20.1	26.8
11	8:40	—	47.95	36.65	21.10	20.80	21.10	—	21.0	26.95
12	9:30	—	49.0	37.9	22.85	22.55	22.85	—	22.85	26.15
13	9:31 am	Rated current	current	applied	14.6A					
14	10 31 am	49.5	38.95	24.9	25.1	24			24.67	
15				24.2	24.1				24.01	
16										

Observation / Result	Top oil Temperature - Rise	26.01
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Customer Signature

Tested by

Test Engineer

Approved BY

DOCUMENT : FORMS & FORMATS
SECTION : 44 PAGE NO.1 of 2ISSUE NO.: 2
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Step 1	☒ After top oil reached steady state temperature, Injection of rated current (<u>14.58</u> A) on Supply side of Transformer under Test for One Hour	
☒	After Centre of top yoke / LV core reached steady state temperature in case of Dry Type after No load Run (or) Load Run _____	
(i) Measurement of Winding Resistance on HV _____ (or) LV _____ side		
Winding material :	Copper ☒ Aluminum ☐	Thermal Co-efficient : 225 ☐ 235 ☒ 234.5 ☐
Time of Shutdown : Correction Factor (if applicable): <u>0.475</u>		
Time in Sec./min.	☐	☒
Resistance in Ω / m Ω	☐	☒
Temperature Rise of HV _____ (or) LV _____ Winding in $K/^\circ C$		
$\frac{4.0464}{3.5540} (235 + 25.9) - 235 = 22.85$$+ 0.475 = 39.7^\circ C$ within limits		
Step -3	(ii) After the measurement of Resistance on HV _____ (or) LV _____ side in above step, rated current was injected for one more hour before taking resistance of leftover HV _____ (or) LV _____ winding	
Time of Shutdown : Correction Factor (if applicable): _____		
Time in Sec./min.	☒	☐
Resistance in Ω / m Ω	☒	☐
Temperature Rise of HV _____ (or) LV _____ Winding in $K/^\circ C$		
$\frac{4.3425}{3.8844} (235 + 25.9) - 235 = 22.85$$+ 0.475 = 34.3^\circ C$ within limits		
Customer Signature _____ Tested by <u>Red</u> Test Engineer		
Instruments / Equipment used	Serial Number	Instruments / Equipment used
Power Analyzer		DTM
Resistance Meter		Data Logger
Timer		
Approved BY _____		
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